🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 2 · STANDARD 6.NOS.1

Fraction Division Problem Solving

Lesson 2-5 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can solve real-world problems by writing and solving fraction division equations.
Language Objective: I can explain my reasoning using the words model, equation, solution, and reasonableness.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Model | **Equation** | **Solution** | **Reasonableness** | **Inverse operations**

👆 *Tap any word to see what it means and a picture.*

1. A drawing or diagram that shows a math situation is a ________.

2. A math sentence that says two amounts are equal, with an = sign, is an ________.

3. The value that makes an equation true is the ________.

4. Checking whether an answer makes sense for the problem is checking its ________.

5. Operations that undo each other, like multiplication and division, are ________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

What clue in a word problem tells you to divide fractions instead of multiply?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Equation** — A math sentence with an equal sign showing both sides are the same.
  Ecuación — Una oración matemática con un signo igual que muestra que ambos lados son iguales.
- ☐ **Solution** — The answer to an equation or problem.
  Solución — La respuesta a una ecuación o problema.
- ☐ **Reasonableness** — Checking if your answer makes sense.
  Razonabilidad — Revisar si tu respuesta tiene sentido.
- ☐ **Model** — A picture or math way to show a problem so you can solve it.
  Modelo — Un dibujo o forma matemática de mostrar un problema para resolverlo.
- ☐ **Inverse operations** — Two math actions that undo each other, like × and ÷.
  Operaciones inversas — Dos operaciones que se deshacen entre sí, como × y ÷.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The problem asked me to find ____, so I divided.
El problema me pidió encontrar ____, así que dividí.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: The total amount is the dividend; the size of each test is the divisor.

Evidence: Students identify 2/3 as the total being split (dividend) and 1/6 as the group size (divisor), writing 2/3 ÷ 1/6.

Reasoning: This evidence connects the definition of model to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The problem asked me to find ____, so I divided.**
  El problema me pidió encontrar ____, así que dividí.
- **I divided rather than multiplied because ____.**
  Dividí en lugar de multiplicar porque ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Model
2. **Notes 2:** Equation
3. **Notes 3:** Solution
4. **Notes 4:** Reasonableness
5. **Notes 5:** Inverse operations
6. **Watch for this mistake:** *A common mistake in Fraction Division Problem Solving is reversing which fraction is the dividend and which is the divisor when setting up the equation from a word problem. For example: a pitcher holds 3/4 cup of punch, and each serving is 1/8 cup — how many servings? The correct setup is total ÷ group size = 3/4 ÷ 1/8 = 3/4 × 8/1 = 24/4 = 6 servings. A student who reverses the setup calculates 1/8 ÷ 3/4 = 1/8 × 4/3 = 4/24 = 1/6, an answer smaller than 1 that can't possibly be a serving count. Before you submit, check: does my answer make sense as a NUMBER OF GROUPS? If the setup gives a fraction less than 1 for a 'how many' question, you likely swapped the dividend and divisor.*
7. **Try It 1:** A. 5 — *The total distance (5/6 mile) is the dividend and the checkpoint spacing (1/6 mile) is the divisor: 5/6 ÷ 1/6 = 5/6 × 6/1 = 30/6 = 5 checkpoints.*
8. **Try It 2:** A. 3 — *The total (9/10 liter) is the dividend and each vial (3/10 liter) is the divisor: 9/10 ÷ 3/10 = 9/10 × 10/3 = 90/30 = 3 vials, so the code is 3.*